

HealthConnect Clinic

Week 5 Project Summary

AnalystLab Africa Experience Lab

Track: Data Science

Week 5 Objective

The goal for Week 5 was to move the HealthConnect project from problem definition into practical machine learning development.

The main focus was data preparation, exploratory analysis, feature engineering, and the development of an initial model for predicting appointment no-shows.

Work Completed

The appointment dataset was cleaned and prepared for modelling.

Missing reminder channels were labelled as No Reminder, while missing distance and waiting-time values were handled using median imputation.

Cancelled appointments were excluded because the prediction problem focuses on Attended vs No-Show appointments.

EDA was completed to examine the target distribution and relationships between patient, booking, reminder, and appointment variables.

Several new features were explored. These included previous no-show rate, previous no-show history, booking lead information, and booking month.

A patient-level train/test strategy was used to prevent appointments belonging to the same patient from appearing in both datasets.

Logistic Regression was then developed as the baseline classification model.

Key Results

The baseline model achieved:

Metric	Result
Accuracy	63%
No-Show Precision	62%
No-Show Recall	65%
No-Show F1-Score	64%
ROC-AUC	0.68

The model correctly identified 315 no-show appointments.

Booking lead time showed one of the clearest differences between attended and no-show appointments during EDA.

The results suggest that the available appointment data contains useful predictive information, although the baseline still has considerable room for improvement.

Important Decisions

Cancelled appointments were excluded to keep the prediction problem focused on no-shows.

Patient-level splitting was selected instead of a normal random row split. This prevented the same patient from appearing in both training and testing data.

Highly redundant engineered features were not all retained in the baseline model.

Logistic Regression was selected because it provides a simple and interpretable starting point for future model comparison.

Challenges

Some variables contained missing information and required appropriate treatment.

Repeated appointments from the same patients also created a potential leakage risk.

Several predictors showed weak individual relationships with appointment outcome, which limited the predictive strength of the initial model.

Changes From Week 4

Week 4 focused mainly on planning the proposed machine learning approach.

Week 5 tested those decisions using the actual data.

The initial feature set was refined after EDA and correlation analysis. A patient-level splitting strategy was also implemented to provide a more realistic evaluation.

Cross-Track Collaboration

I collaborated with a Data Analytics intern to compare findings related to appointment attendance.

Their analysis highlighted reminder status, previous no-show history, and booking lead time as relevant factors.

These findings supported patterns observed during my Data Science analysis and strengthened the interpretation of the selected predictors.

Remaining Work

The current Logistic Regression model is an initial baseline rather than a final solution.

Further work is required to compare alternative algorithms, improve feature selection, examine model interpretation, and evaluate whether performance can be improved.

Week 6 Focus

Week 6 should build directly on the current baseline.

The next stage will focus on:

- Comparing additional classification models
- Improving feature engineering and selection
- Evaluating model performance
- Exploring model interpretation
- Reviewing decision thresholds
- Assessing potential fairness concerns
- Comparing improved models against the Week 5 baseline

Week 5 Status: Baseline model successfully established.